SRSG and S2SG: A Model and a Dataset for Scene Graph Generation of Remote Sensing Images From Segmentation Results

Zhiyuan Lin, Feng Zhu, Yanzi Kong, Qun Wang, and Jianyu Wang

***Abstract*— Remote sensing image analysis has drawn more attentions in the field of computer vision. At present, the methods commonly used in remote sensing image analysis are mainly at a low level semantically. The scene graph is an abstraction of objects and their relationships, which is a high-level image understanding task. To fully comprehend the meanings of remote sensing images, in this article, we propose a novel segmentation-based model to generate remote sensing image scene graphs (SRSG). In the SRSG model, a more complete and accurate scene graph is generated with the segmentation results as inputs, while the shapes of objects are reasonably coded. The morphological features of object pairs are embedded together by different branches of the SRSG model and then mapped to semantic space to predicate their relationships. Furthermore, a new dataset for scene graph generation of remote sensing images, namely, segmentation results to scene graphs (S2SG), is constructed based on pixel-level segmentation results. Experimental results demonstrate that the performance of the SRSG model is far superior to the previous methods in the task of generating remote sensing image scene graphs. The proposed SRSG model opens up new possibilities for remote sensing image analysis at a high level. Moreover, the S2SG dataset further allows for the evaluation of different approaches and is provided for the benefit of the research community.**

***Index Terms*— Dataset, relationship prediction, remote sensing image, scene graph generation, segmentation.**

I. Introduction

COMPUTER vision has got rapid development in recent years, mainly due to the explosive growth of image dataset. Noticeable progress has been made in lots of computer vision applications. These applications can be divided into two categories according to the level of semantics: image processing focusing on individual instances and the comprehensive understanding of image content. The former includes object detection [1]–[4], semantic segmentation [5], [6], and instance segmentation [7], [8], and the latter involves image captioning [9], [10], visual question answering [11], [12], and visual reasoning [13]. Many studies have begun to explore the relationships between objects, aiming at understanding the images' contents semantically. The scene graph [14]–[17] is a structural description of an image and a bridge between low-level detection and high-level cognition [18], the nodes and edges of which represent the detected objects and semantic relationships between instances, respectively. The scene graph generation is an increasingly important area in natural image understanding.

However, it is worth noting that the technological advancement of remote sensing image understanding based on computer vision is far behind natural images. The researches to date have tended to focus on primary applications in remote sensing images, such as object detection [19] and semantic segmentation [20]. These works are not enough to support high-level tasks, such as decision making. As a visual task with a slightly higher semantic level than primary applications, scene graph generation plays a significant role in remote sensing images understanding, but there are few related studies at present. Therefore, in this article, we conduct preliminary research on the scene graph generation of remote sensing images, including the segmentation-based model to generate remote sensing image scene graphs (SRSG) and the dataset called segmentation results to scene graphs (S2SG).

Scene graph generation is one of the basic methods to extract the semantic content of images. Its function is to represent the image as a graph. As shown in Fig. 1(c), the scene graph is composed of a series of triples of the form ⟨subject-predicate-object⟩. The nodes and edges in the scene graph represent the object pairs ⟨subject, object⟩ in the image and the predicate between them, respectively. Predicting reasonable predicate is the critical process of scene graph generation while given the object pairs ⟨subject, object⟩. It is essentially the process of mapping the features of object pairs ⟨subject, object⟩ into semantic space that contains all predicates. The scene graph generation could be sorted into three subtasks: predicate classification, phase classification, and scene graph generation. Given the ground truth locations and categories of the objects in the image, the first subtask is to predict the

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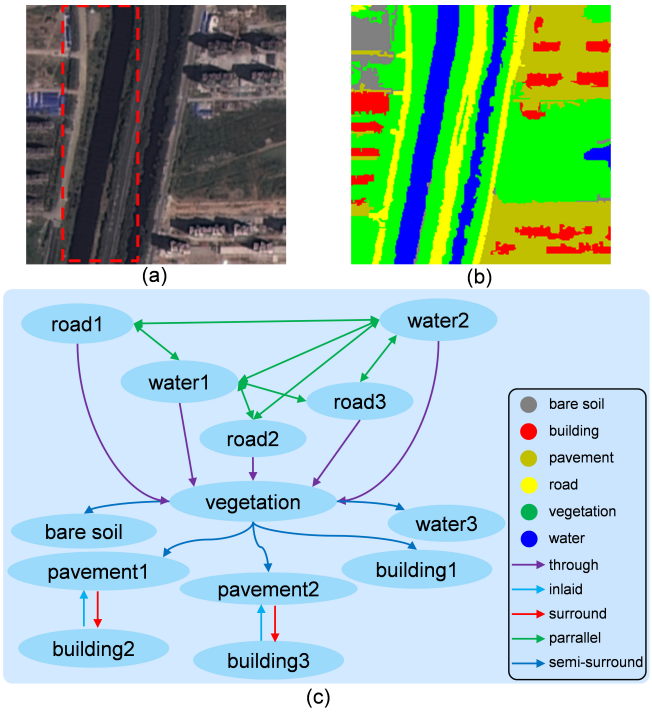


Fig. 1. (a) Visible light remote sensing image. The red dashed box is the bounding box of the river. (b) Segmentation result of image (a). (c) Scene graph corresponding to this remote sensing image. The nodes in the scene graph correspond to each object in the segmentation results, and different arrows correspond to different predicates.

relationship between two instances. The second subtask is to predict the classes and the predicates in object pairs, given the ground truth locations of instances in the image. The objects in the image and the relationships between them are detected by the third subtask, simultaneously [21]–[23].

Almost all the existing scene graph generation methods perform on natural images. However, due to the differences between remote sensing images and natural images, the existing methods are difficult to complete the task of remote sensing image scene graph generation. The object in natural images mainly concentrates in an area and can be marked by a bounding box without introducing noises. However, a single object in remote sensing images may have a large scale and spread across the entire image [24]. As shown in Fig. 1, the river passes through the whole scene thoroughly; the vegetation fills the entire image. In this case, the proportion of background in the bounding box is even more than that of the object itself, such as the labeled river in Fig. 1(a), so it is difficult for the bounding boxes to mark remote sensing image objects accurately. Another characteristic of remote sensing images is that the objects are highly mixed. As shown in Fig. 1(b), water 1, water 2, and all roads extend from one side of the vegetation to the another, so the relationship between these instances and the vegetation can be regarded as "through": ⟨water-through-vegetation⟩. The building and pavement are mixed together, so they can be summarized as the following triples: ⟨pavement-surround-building⟩ and ⟨building-inlaid-pavement⟩. In addition, we found that the corresponding predicates will be different in remote sensing images after interchanging the subject and the object, such as the "pavement" and "building" in Fig. 1(c). Different predicates are indicated by various arrows in the example of Fig. 1(c). The double-headed arrows indicate that even if the subject and object are interchanged, there is still an actual semantic relationship between them. This is also very different from natural images; the subject and object in natural scene graphs are rarely interchangeable.

After comparing the natural scene images and remote sensing images, we find that the relationships between objects in remote sensing images depend more on their relative spatial positions and morphology. Therefore, the relationships between objects in remote sensing images can be predicted only by strictly labeling the objects, and high-quality scene graphs can be constructed only by extracting the objects' morphological features. The method that locating objects with bounding boxes in natural images cannot accurately label the objects in remote sensing images. The bounding box also fails to include the morphological features of the objects. Based on this, we creatively build the SRSG model, which takes the pixel coordinates of the segmentation results as an input to encode the objects' morphological features and relative spatial positions and predict the predicates between them. To the best of our knowledge, this is the first work to do so.

There are almost no datasets with semantic relationships annotated in detail for remote sensing image scene graph generation. There is a high cost of time and labor to construct a large and detailed annotated dataset [25], [26]. The scene graph dataset with detailed instance relationships is even worse. Because multiple predicates can be used to express the relationship of an object pair, some scenes also lack strict definitions, making the search space of the scene graph generation task more immense and its difficulty significantly increased [21]. The lack of datasets severely limits the research on scene graph generation for remote sensing images. From the above analysis, a remote sensing image scene graph generation dataset based on segmentation results is constructed, called S2SG. In S2SG, each object is represented by an instance segmentation result instead of a bounding box, and the relationships between objects are represented by appropriate predicates. Since the relationships between the objects in remote sensing images are dependent more on their shape and mutual morphological distribution, in S2SG, the coordinates of each pixel in segmentation results are reserved for coding the object shape while the visual features are discarded.

The experimental results show that accuracy and complete scene graphs of remote sensing images will be generated by SRSG trained on S2SG. In summary, the contributions of our work are fourfold.

1) We have analyzed the features of remote sensing images and attempted the scene graph generation of remote sensing images based on the pixel-level segmentation results for the first time.
2) A dataset S2SG is constructed to advance the task of remote sensing image understanding.

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3) We build the SRSG model that takes the segmentation results as inputs to generate remote sensing image scene graphs.
4) The SRSG model is trained and tested on the dataset S2SG and outperforms previous work.

Our work provides new ideas for the study of remote sensing image understanding. The specific implementation of the model and the dataset will be described in detail in Section III.

II. Related Works

There have been many achievements in scene graph generation research on natural images. Visual Genome [27] proposed a dataset, which marks the relationship between objects, the attributes of the objects, and the description of the scene. This dataset is one of the essential materials in the scene graph generation work. Lu *et al.* [21] constructed the scene graph generation dataset for visual relationship detection (VRD) and predicted the visual relationship by language priors. Xu *et al.* [14] predicted the relationship between objects by contextual message passing. Tang *et al.* [28] enhanced the relationship between the objects and the contents of the entire image by a dynamic tree structure. The scene graph is generated from three dimensions of objects, phrases, and region captions by a dynamic graph in [29]. Union Visual Translation Embedding (UVTransE) [30] indicated that predicting the relationship between objects is not always straightforward visually and introduced contextual information to assist scene graph generation.

Reasonable nonvisual information is crucial for generating high-quality scene graphs. The image-agnostic feature is defined in [26]. It is pointed out that the relationship between objects mostly depends on the position of the object in the image and the relative position of the two entities. Similarly, Zellers *et al.* [15] found that the object labels can predict many relationships in the scene graphs without excessive visual features. Yu *et al.* [31] introduced that the external textual knowledge to the network significantly enhanced the flexibility of the generated scene graph. Scene graph generation also faces some typical problems, such as the long-tail distribution of the existing datasets. The scene graphs generated by the existing methods weaken the gist of the images [32]. These problems severely reduce the flexibility of scene graph generation. Tang *et al.* [22] used counterfactual causality to eliminate bad bias and increase the flexibility of scene graph generation. Hierarchical Graph Attention Network (HGAT) [33] captured the dependencies among relation triplets by triplet-level graphs, enhancing the hierarchy of scene graphs.

Much research has been done on scene graph generation for natural images, but few researchers focus on understanding remote sensing images semantically. Shi and Zou [34] explored whether the machine can automatically generate humanlike language descriptions of remote sensing images and proposed a framework for remote sensing image captioning. This work makes a preliminary exploration of the high-level understanding of remote sensing images. Truncation Cross Entropy (TCE) loss [35] is proposed to alleviate the overfitting problem of remote sensing image captioning. Considering the characteristics of remote sensing images, the Remote Sensing Image Captioning Dataset (RSICD) [24] is constructed to generate more vivid remote sensing image descriptions. A multiscale remote sensing image interpretation network (MSRIN) [36] to identify the spatial relationship of objects in the remote sensing image. Still, the spatial relationship between objects is only included in the image's descriptions and is not displayed explicitly. Li *et al.* [23] explored the generation of remote sensing image scene graphs and proposed a remote sensing image scene graph dataset (RSSGD), in which bounding boxes are used to label the objects. This research integrates global information by graph convolutional networks to improve the accuracy of scene graph generation, but it does not profoundly analyze the characteristics of remote sensing images. The influence of objects' morphology on the scene graph of remote sensing images is ignored.

It can be seen that scene graph generation is one of the demonstrations of scene understanding that thoroughly abstracts the scene contents. It has also achieved some results on natural scenes, but the research on remote sensing images is far from enough. On this basis, we launched the study of this article.

III. Dataset and Method

The dataset (S2SG) and model (SRSG) for remote sensing image scene graph generation will be introduced in this section in detail.

A. Segmentation Results to Scene Graphs

Datasets are one of the primary supports for the development of data-driven approaches. The slow progress of remote sensing image research based on deep learning is partly due to the lack of large datasets in this field. There are almost no datasets about remote sensing image scene graph generation. Therefore, in this work, the dataset S2SG is first constructed based on the characteristics of remote sensing images.

The predicates that reasonably represent the relationships between objects in remote sensing images are selected first. The remote sensing image's contents are abstracted in the captions, so the words in the captions are an excellent corpus for selecting predicates. We select predicates from three remote sensing image description datasets, RSICD [24], Sydney-captions [37], and University of California at Merced (UCM)-captions [37]. There are nearly 68 000 descriptive sentences about different remote sensing scenes in the three datasets. Next, we perform statistics on the frequency of each kind of predicates in these three datasets. Predicates with higher frequencies represent more common semantic relationships in remote sensing images. The 80 predicates with the highest frequency are selected for the following selection. Then, we calculate the semantic distance of these predicates and remove words with significant semantic overlap. The four orientation words "south," "east," "north," and "west" are removed because the pose of the image acquisition device is unknown, and the genuine orientation relationship cannot be represented by the orientation words either. The four words "top," "below," "left," and "right" are reserved to

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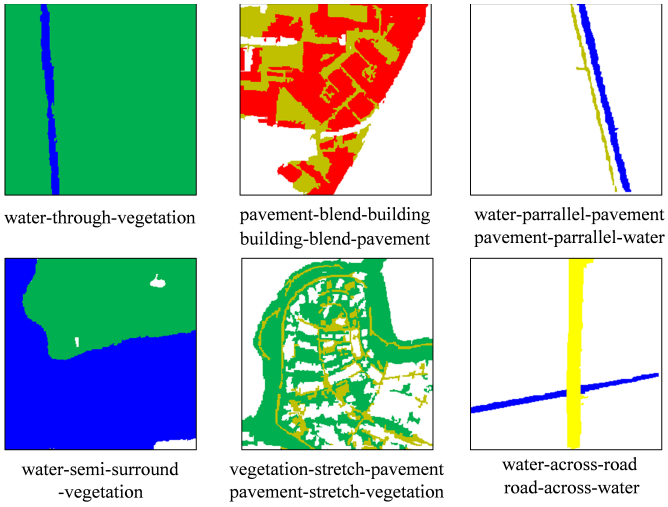


Fig. 2. Graphical representation of some relationships. It can be seen that some predicates can adapt to the interchange of subject and object, such as "blend," but others are not, such as "through" and "semisurround."

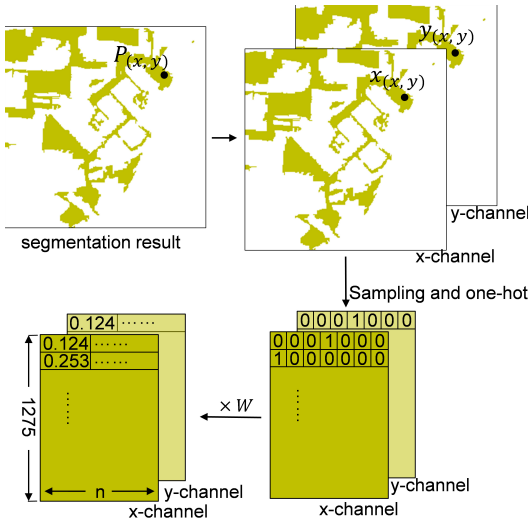


Fig. 3. Coordinates of each position in an object segmentation result are encoded by the process. In this way, the different morphological patterns of the objects can be uniformly expressed.

indicate the relative position of the objects in the scene. After screening, 12 predicates expressing the relationship between objects in remote sensing images are retained as follows: "through," "stretch," "surround," "across," "top," "blend," "parallel," "inlaid," "right," "left," "below," and "semisurround." Most of the high-frequency and more typical relationships in remote sensing images can be expressed by the 12 predicates. Fig. 2 shows the visual display of some relationships. These relationships mainly depend on the morphology of the two adjacent objects, and visually, some relationships in remote sensing images are more complicated than those in the natural scene. One thing to note is that in S2SG, we only define the relationship between objects immediately adjacent to each other.

After determining the predicates used to represent relationships, it is essential to construct the complete dataset with appropriate remote sensing images. Because the objects in remote sensing images are mostly all over the image or mixed, the area marked by the bounding box contains many messy features. The objects cannot be accurately labeled in this way. However, the objects can be accurately distinguished along with their edges by pixel-level segmentation. This characteristic makes it possible to generate high-quality scene graphs for remote sensing images. Therefore, based on the remote sensing image dataset WuHan Dense Labeling Dataset (WHDLD) [38], a small dataset of remote sensing image scene graph generation based on segmentation results S2SG is constructed. In the dataset, 12 relationships are annotated in detail, and six classes of objects are remained, namely, bare soil, building, pavement, road, vegetation, and water.

Some datasets, whether for object detection or scene graph generation, contain thousands of images. However, even if these datasets contain more images, they cannot effectively avoid long-tail distribution because the frequency of each object category in images is different, as in the real world. The proportion between categories in the real world is different, which will be reflected in datasets. The prediction of the relationship between objects faces the same problem. Although there are some studies on hard examples [39], designing a balanced dataset is the key to learn each relationship thoroughly.

The proportion of the relationship between objects in the training set will affect the model's performance. To avoid the long-tail distribution, the number of each relationship should be consistent. However, the number of relationships between objects in each training image is different, contradicting our expectations. The contradiction is because the existing datasets use the number of images as the unit of dataset size measurement. Each image is densely annotated without maintaining the uniformity of the distribution of different relationships. So, in S2SG, the count of each relationship is used to measure the dataset size, and the image number contained in the dataset is ignored. The training set of S2SG contains a total of 1200 object relationships, and the predicate of each relationship corresponds to 100 training examples. Not all objects are intensively annotated in the training set, and only object pairs with apparent relationships are marked. In the test set, the relationship between all objects of each image is marked in detail and densely. The model can be trained unbiased on the dataset designed in this way. Because only significant object relationships are marked in the training set, it is difficult for the model to learn the object relationships that do exist in an image but are not marked. These relationships can only be learned in other labeled images. Therefore, we have modified the input of our model SRSG accordingly to adapt to this dataset form, and the implementation method will be introduced in the experiment.

B. SRSG

The semantic relationships of objects in remote sensing images are expressed mainly by the morphological pattern and relative spatial positions between objects. Most of them are spatial relationships, such as up and down, left and right, through, surrounding, and so on. The key is how to encode the relative spatial position relationships and the architecture

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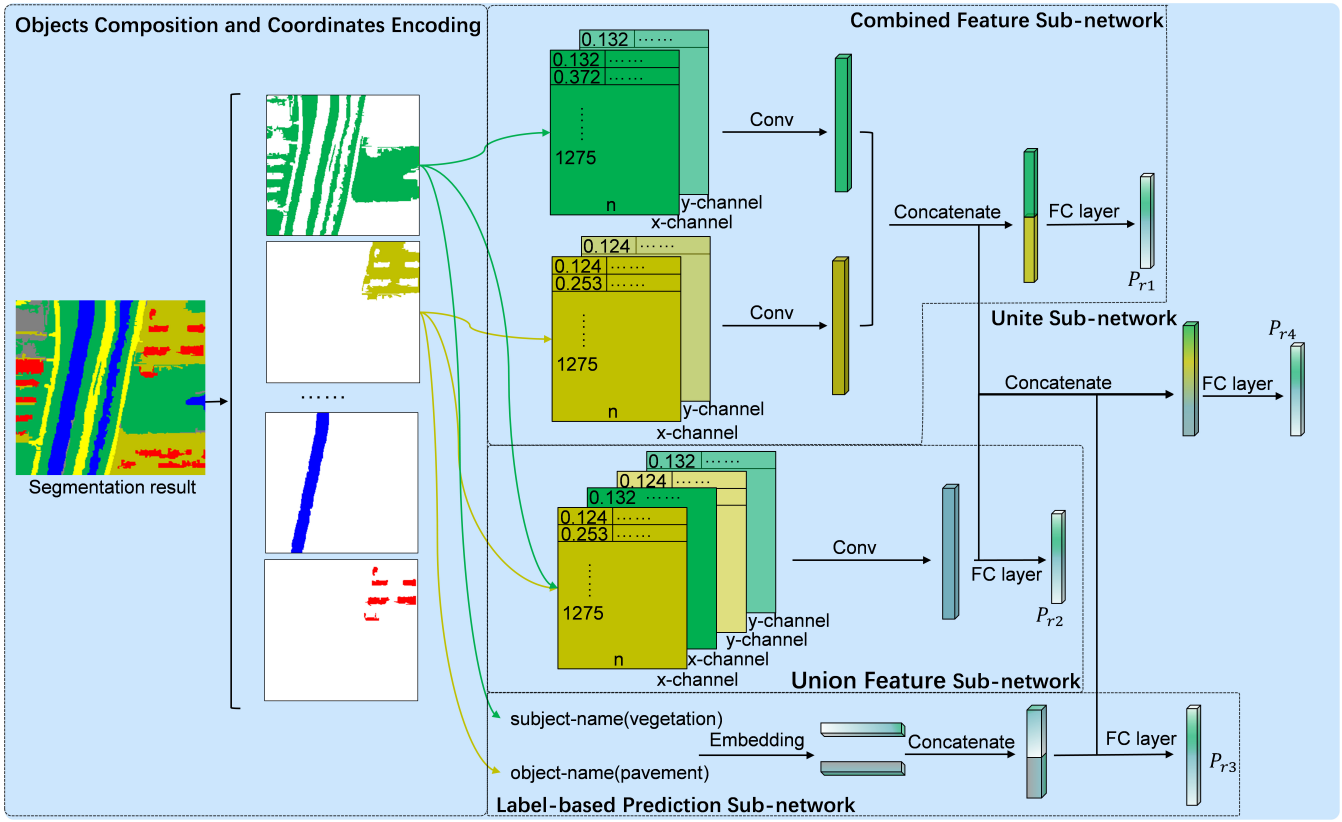


Fig. 4. Illustration of the SRSG model. The input of the SRSG model is the pixel-level segmentation of remote sensing images. When predicting the relationships, the different objects in the segmentation result are encoded first to form the unified inputs, which is the process shown by the yellow and green arrows. P_{r*} represents the predicted probability of the predicates for different branches. During training, the loss function integrates the four branches' output, and P_{r4} is used as the final prediction result.

information of object pairs. The position coordinates of object pairs implicitly contain both kinds of information. Therefore, we encode the coordinates of each object pair and input the SRSG. Moreover, image-agnostic features such as object labels can predict the relationship between objects [15], [26]. So, we also take the labels of object pairs as abstract semantic representations of their visual content to input to the SRSG.

It is different from other methods that input the bounding box into the model. In the SRSG model, the instances' coordinates in the image segmentation results are encoded first. Then, the encoded coordinates and the labels of subject and object that in an instance pair are used as an input. There are four branches in SRSG that combine and process the input features of subjects and objects. The model will output the probabilities of the predicates representing the semantic relationships between subjects and objects.

1) Encoding the Coordinates of Instances: Because we use a segmentation-based method to predict the relationship between two adjacent objects by the morphological distribution, the morphology of the two instances should be normalized before inputting the model. Before training, the visual features of each object in segmentation results are discarded, and every pixel is replaced by the position coordinate of this pixel, that is

$$\text{pix}_{i(x_i,y_i)} \rightarrow \{x_{i(x_i,y_i)}, y_{i(x_i,y_i)}\}, \quad i \in \boldsymbol{P} \tag{1}$$

where $\boldsymbol{P}$ is the point set of object P segmentation result, i is the ith point in it, $\text{pix}_{i(x_i,y_i)}$ is the pixel value at (x_i, y_i), and (x_i, y_i) is the coordinate of i. The pixel at this position becomes two values, so we accommodate the extra value by expanding the channel to form an x-channel and a y-channel. Fig. 3 shows the changes.

The absolute size of the coordinate itself causes the above conversion in (1) not to represent the shape of an object well, so it is necessary to encode the value of the coordinate itself to achieve its normalization. Since the resolution of the image is 256 × 256, it only needs to encode 0–255. The embedding method in natural language processing is adopted as follows:

$$\boldsymbol{x}' = F_{\text{one-hot}}(x) \cdot W \tag{2}$$

where x is the value to be encoded, and $\boldsymbol{x}'$ is the encoding vector of x. In (2), x is first changed into its one-hot form, and then, it is mapped to $\boldsymbol{x}'$ by the learnable parameter W. In this way, with the training of the model, both the x- and y-coordinates will become a vector of length$n$, where n is a hyperparameter.

Since the pixel number in the segmentation results of each object is different, the number of coordinate points contained will not be consistent, and the coded object cannot be directly input into the model. So, the encoded object is flattened and performed 1-D pyramid sampling, so that each instance seg-

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mentation result can be expressed by a tensor of $2 \times 1275 \times n$, where 2 refers to two coordinate channels, the x-channel and y-channel, 1275 is the number of sampling points for each segmented object, and n is the length of the vector for each real number. In this way, the standardization of the input data is realized. Fig. 3 shows the preprocessing of encoding each position coordinates in an object segmentation result.

2) Structure of SRSG: The SRSG model is combined with four subnetworks. The first two subnetworks extract the subject and object features in an object pair ⟨subject, object⟩ and combine them in different ways. The third subnetwork takes the labels of subject and object as an input to provide the semantic information of image content for the model. The fourth subnetwork integrates the features produced by the first three subnetworks to predict the predicate for the current object pair finally. The structure of the four subnetworks is shown in Fig. 4.

As shown in the P_{r1} branch in Fig. 4, called the combined feature subnetwork, the encoded coordinates of the subject and object are first convolved and linearly activated in the first subnetwork branch. Then, the subject and object features are concatenated into combined features to predict the predicate corresponding to the relationship between the instances. The whole process can be expressed as follows:

$$\text{softmax}\left(L\left[L\left(\text{Conv}\left(D_{\text{subject}}\right)\right), L\left(\text{Conv}\left(D_{\text{object}}\right)\right)\right]\right) \rightarrow P_{r1} \tag{3}$$

where D_{subject} and D_{object} represent the encoded coordinates of the subject and object, respectively. Conv and L represent the convolutional layer and the fully connected layer, respectively. [] means to connect two feature vectors together. P_{r1} represents the probability of the predicate for the current instance pair.

The P_{r2} branch in Fig. 4 shows the structure of the union feature subnetwork. Unlike the P_{r1} subnetwork, the subject and object features are first combined to form a tensor and then fed into the convolutional and the fully connected layer. The tensor size is $4 \times 1275 \times n$, where 4 refers to the x-channel and y-channel of the subject and the object, a total of four coordinate channels. The union feature subnetwork finally outputs the predicate's probability P_{r2} corresponding to the subject and object relationship. That is, the encoded morphological features of the subject and object are connected first in the channel dimension, and then, the relationship category is predicted. It can be expressed as follows:

$$\text{softmax}(L(\text{Conv}\left[D_{\text{subject}}, D_{\text{object}}\right])) \rightarrow P_{r2}. \tag{4}$$

The P_{r3} branch in Fig. 4 shows the label-based prediction subnetwork. The subnetwork takes the subject and object labels as an input. The labels are encoded first by the word embedding to obtain their word vector. Then, the two word vectors are connected and input to the fully connected layer. Finally, the fully connected layer outputs the predicted probability P_{r3} of the predicate corresponding to the relationship between the objects. The process is as follows:

$$\text{softmax}(L\left[E(\text{Sub}_{\text{label}}), E(\text{Ob}_{\text{label}})\right]) \rightarrow P_{r3} \tag{5}$$

where E means the word embedding process of the label, and the method is the same as (2). This subnetwork aims to predict the relationship between objects by labels because some relationships are inherent to some specific objects [15]. Moreover, the object labels as image-agnostic features provide semantic knowledge for predicting. The object labels are also a semantic representation of the image content. This is also one of the motivations for many work on scene graph generation with external knowledge [16], [31].

After constructing the three subnetworks, the unite subnetwork is used to integrate the three subnetworks into one, as shown in the P_{r4} branch. In the unite subnetwork, the convolutional features of subject and object: $\text{Conv}(D_{\text{subject}})$ and $\text{Conv}(D_{\text{object}})$, the union features of subject and object: $\text{Conv}[D_{\text{subject}}, D_{\text{object}}]$, and their label features: $[E(\text{Sub}_{\text{label}}), E(\text{Ob}_{\text{label}})]$, are connected to form comprehensive features. Then, the comprehensive features are fed into the fully connected layer to predict the predicate. The whole structure is as follows:

$$\text{softmax}\left(L\begin{bmatrix}\text{Conv}\left(D_{\text{subject}}\right), \text{Conv}\left(D_{\text{object}}\right)\\ \text{Conv}\left[D_{\text{subject}}, D_{\text{object}}\right]\\ \left[E(\text{Sub}_{\text{label}}), E(\text{Ob}_{\text{label}})\right]\end{bmatrix}\right) \rightarrow P_{r4}. \tag{6}$$

After inputting the comprehensive features, the predicate between the current objects will finally be predicted by the unite subnetwork.

The first three subnetworks can calculate the predicate probability and loss of the corresponding branch. The unite subnetwork can calculate the comprehensive loss of the entire network. Therefore, the final loss function of the network is expressed as

$$\text{loss} = W_{\text{co}} \cdot L_{\text{vec}} = \alpha \cdot l_{\text{com}} + \beta \cdot l_{\text{union}} + \gamma \cdot l_{\text{label}} + \varphi \cdot l_{\text{unite}} \tag{7}$$

where l_{com}, l_{union}, l_{label}, and l_{unite} are the cross entropy losses of the four subnetworks, and finally, they are combined into the total loss. W_{co} represents the coefficient matrix of the loss function. The model will be able to extract global features more efficiently by calculating the loss of each subnetwork. It can also indirectly verify which combination has the most significant impact on generating remote sensing image scene graphs. Such a structure enables the network to converge quickly, and different input combinations will enhance the network's performance. The result of different combinations and hyperparameter selection will be analyzed in detail in Section IV.

IV. Experiment and Discussion

The dataset S2SG, the format of input data, and the SRSG model have been detailed in Section III. How to train the SRSG model on the S2SG dataset and generate remote sensing image scene graphs based on the pixel-level segmentation will be introduced in this section.

A. Training and Prediction Process

In the dataset S2SG, to avoid the long-tail distribution caused by the different numbers of relationships in each image,

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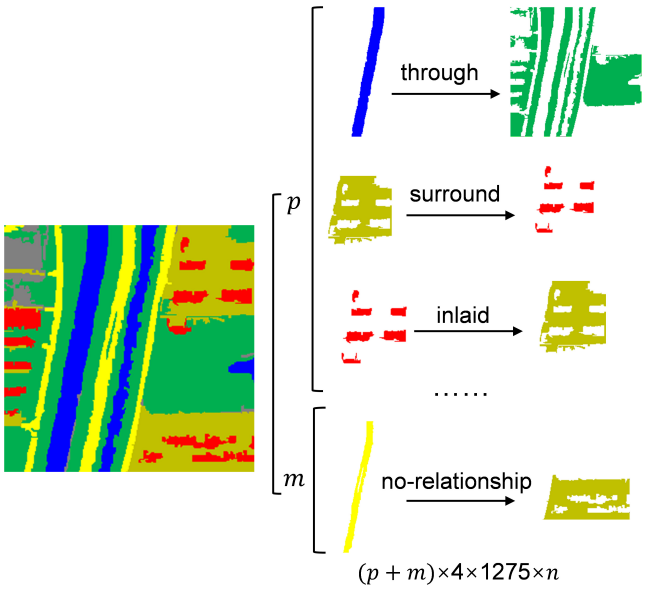


Fig. 5. Process of decomposing the segmentation result into the input of the SRSG model. $(p+m) \times 4 \times 1275 \times n$, respectively, correspond to the four dimensions of batch size, channel, and resolution required in the model.

we no longer use the number of images to measure the size of the dataset. In this dataset, the number of relationships is used to estimate the dataset's scale, and the number of each relationship is kept consistent; 12 kinds of relationships contain a total of 1200 examples. According to the characteristics of the dataset S2SG, we have also made corresponding adjustments when training the model.

The structure of SRSG makes it impossible to input the entire image for training directly. The segmentation result of an image will be decomposed into several object pairs as an input to the model. If a segmentation map contains q objects, then $q \times (q-1)$ object pairs can be formed. Suppose p object pairs are marked with actual semantic predicates and regarded as positive samples, then $q \times (q-1) - p$ object pairs that do not have actual semantic relationships are considered to be negative samples. During training, p positive samples and randomly selected m negative samples are used as a batch, namely, $(p+m)$ in Fig. 5 is input the model for training. It is equivalent to include $(p+m)$ examples in an image. Therefore, each image is transformed to a tensor with the scale of $(p+m) \times 4 \times 1275 \times n$, and the whole process is shown in Fig. 5. The composition of each training sample of SRSG includes three parts: the coordinates of regions corresponding to the subject and object in a remote sensing image and the predicates between them. The coordinates of the two areas are input to SRSG to extract their morphological features, and the predicate is used to compare with the model's output and calculate the loss. Even if the number of examples contained in each image is different, it can be used to train the model. The balance of positive and negative examples is maintained by adjusting m, so that the model can be trained better. After inputting the coordinates of the regions where the two objects are located, the probability of the predicate between them will be calculated by the SRSG model. This process is essentially a multiclass classification problem. Here, we use the cross entropy loss as the loss function (7).

The $q \times (q-1)$ object pairs consisting of all instances in an image are input into the SRSG model when predicting. Then, the model will output the probability of the relational predicate category between the objects. Finally, a scene graph of the remote sensing image is generated.

TABLE I

PERFORMANCE OF THE SRSG MODEL WHEN USING DIFFERENT SUBNETWORKS AND LOSS FUNCTIONS

Sub-network			loss		mR@50	R@50	mR@20	R@20
s_{com}	s_{union}	s_{label}	l_{unite}	l_{all}				
√			√		0.1166	0.1694	0.0622	0.1103
√	√		√		0.1313	0.1802	0.0772	0.1183
√	√	√	√		0.2315	0.2121	0.1434	0.1230
√	√	√		√	**0.2561**	**0.3115**	**0.1602**	**0.2330**

B. Experiment and Analysis

In this section, the proposed dataset and the effectiveness of our model will be tested. We make the model more expressive by choosing appropriate hyperparameters. All the experiments were accomplished on a computer with an RTX 2080Ti GPU and a 2.3-GHz CPU. All programs are performed on Ubuntu 16.04 via Pytorch1.10 and Python3.6.

In the experiment, the Recall@K (R@K) [21] and Mean R@K (mR@K) [28], [40] are adopted as the evaluation metrics of the algorithm. After calculating the confidence of all object pairs, the ratio of the correct predicted relationship among the top K confident predictions is calculated to obtain the R@K. MR@K is the average of each category predicate R@K that is calculated on the entire dataset. This allows the model to consider each type of relationship in the dataset without biasing the evaluation due to high-frequency relationships. In our dataset S2SG, the number of each relationship type is used to measure the dataset size, which slows down the quantitative difference between each type, so the R@K and mR@K correspond to the model trained on S2SG should be close.

1) Ablation Studies: Each subnetwork is thoroughly analyzed in this section, and some hyperparameters are reasonably set to make the model achieve the best performance. The specific experiment result is described below.

Table I shows the contribution of each subnetwork to the model. s_{com}, s_{union}, and s_{label} represent the first three subnetworks in Fig. 4, respectively. The marked subnetwork indicates that the subnetwork branch is integrated into the fourth subnetwork branch by function (6). l_{unite} means that only the output of the fourth subnetwork is used to calculate the loss when training the network; that is, α, β, and γ in function (7) are 0, and φ is 1. l_{all} means that the output of the four subnetworks is involved in calculating the loss when training the network, that is, α, β, γ, and φ in function (7) are 1. By carefully

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TABLE II
TEST RESULT OF THE MODEL WHEN DIFFERENT NUMBERS OF NEGATIVE SAMPLES ARE USED

Ng-num	mR@50	R@50	mR@20	R@20
1	0.2439	0.2968	0.1546	0.2120
2	**0.2561**	0.3115	**0.1602**	0.2330
3	0.2390	0.3824	0.1506	0.2922

analyzing the experimental data, we can find that: when more subnetworks are integrated, the mR@K and R@K have been improved significantly. That is to say, after merging more subnetworks, the morphological characteristics of the objects in the remote sensing images can be more fully extracted by the SRSG model, thereby obtaining better performance. When the losses of all subnetworks are integrated into the loss of the model, the model's performance has been further improved, which shows that constraining each branch can make it more targeted to obtain object features from a different perspective. Under this setting, the time for all samples to be input to the model for training, that is, the time for one epoch, is 4.2 s. During testing, the time for generating the scene graph of an image is 41.87 ms on average.

The SRSG model can set the number of object pairs with no relationship in each image before training. The design that dynamically adjusts the percentage of positive and negative examples enables the model to extract the characteristics of the dataset fully and reduces the impact of data bias. The test results of the model under different positive and negative example ratio settings are shown in Table II. The model gets the best performance when the number of object pairs without a relationship in each image is set to 2.

In the above two experiments, the values of the two metrics R@K and mR@K in Tables I and II are close to each other when the mR@K get the highest value. This is because the number of relationships is used to measure the dataset size, and the number of different relationships is kept close in the dataset S2SG. This also proves that the SRSG model pays more attention to comprehensively fitting various types of relationships rather than only learning the characteristics of high-frequency categories.

To investigate the influence of the accuracy of remote sensing image segmentation results on the SRSG model, we use the segmentation maps generated by manual and segmentation algorithms to test the SRSG model. Moreover, different remote sensing image datasets are used to extend our S2SG dataset to validate the transferability of the SRSG model. The detailed data are presented in Table III. The S2SG source indicates the remote sensing image sources used to construct the S2SG dataset. WHDLD is the main part of S2SG, and DeepGlobe [41] is used as an extension to validate the transferability of the SRSG model. The *M*/*A* in Table III indicates that manual or segmentation algorithms generate the segmentation results.

We test the SRSG model by segmentation maps annotated by manual and generated by segmentation algorithm Local Attention Network (LANet) [42] to explore the effect of segmentation results on predicting semantic relationships.

TABLE III
EFFECTS OF DIFFERENT REMOTE SENSING IMAGE DATASETS AND THE SEGMENTATION RESULTS OF REMOTE SENSING IMAGES GENERATED IN DIFFERENT WAYS ON THE SRSG MODEL

S2SG-Source	M/A	mR@50	R@50	mR@20	R@20
WHDLD	M	0.2561	0.3115	0.1602	0.2330
	A	0.2531	0.3224	0.1581	0.2458
DeepGlobe	M	0.2438	0.4497	0.1911	0.3721
	A	0.1952	0.4228	0.1657	0.3444

As shown in Table III, the two types of semantic labels have little impact on the model in the WHDLD, with even a slight increase in R@K, while the metrics decrease slightly in DeepGlobe. This is related to the performance of the segmentation algorithm on the two datasets. The semantic segmentation metrics mean Intersection over Union (mIoU) on the two datasets are 65.2 and 57.5, respectively. Better segmentation results are more favorable for predicting the semantic relationships between instances. This is consistent with the effect of object detection results in natural scenes on the scene graph generation model. Our further analysis finds that among the 12 predicates, the correctness of "stretch" and "blend" dropped more. The reason for this is that both rely on more complex instance morphology. It is worth noting that the dropping of a single category also leads to a decrease in the overall mR@K, which is consistent with the mR@K design. The instance boundary in the segmentation map generated by the LANet is more rounded, which reduces the difficulty of detecting simple predicates, such as "surround." This also explains the increase of some metrics in WHDLD.

Unlike WHDLD, the images in DeepGlobe are labeled unbalanced; that is, the number of each semantic relationship in the training set is not intentionally kept equal. In the training set, there are 100 images and 905 predicates. There are 100 images and 586 semantic predicates in the test set. The 12 relational predicates are the same as WHDLD, but the instance categories are urban, agriculture, rangeland, forest, water, barren, and background. Comparing the SRSG results based on different remote sensing images, we find that the mR@20, R@20, and R@50 are higher, and mR@50 is lower in DeepGlobe. This indicates that the model is able to predict the correct relational predicates more accurately in this dataset and generate higher prediction scores for them. It also shows that the model SRSG has better transferability. The results of both datasets also indicate that predicting the semantic relationships between instances of remote sensing images in WHDLD is more challenging. Due to the unbalanced labeling used in DeepGlobe, it can be seen that the R@K is higher than the mR@K, which indirectly indicates that the balanced annotation in S2SG allows the model to mine the semantic relationships between instances fully.

2) Scene Graph Generation Results and Analysis: Given the characteristics of remote sensing images, scene graphs can be directly generated from the segmentation results by the SRSG model. In this part of the experiment, we compare the results of the model SRSG with the previous work and conduct a detailed analysis. The input of the SRSG model is the pixel-

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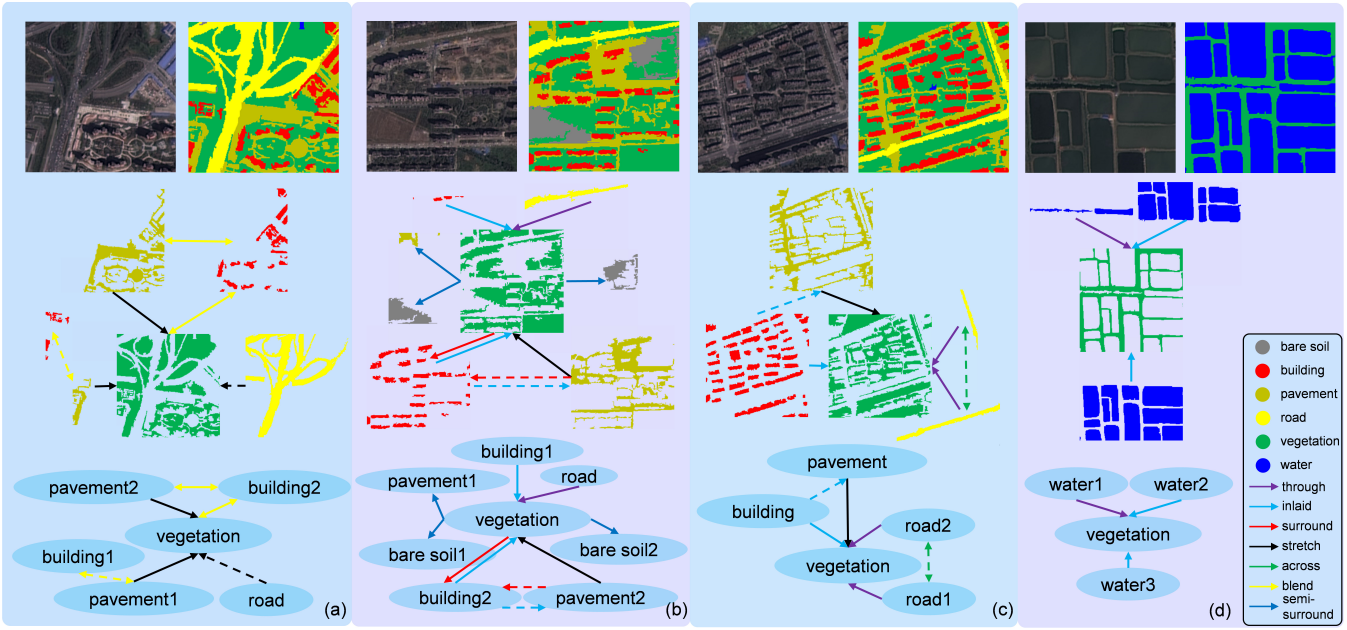


Fig. 6. Visualization of the remote sensing image scene graph generated by the SRSG model. (a)–(d) Corresponding scene graphs generated by the SRSG model when different remote sensing images are fed. The left of the first row of each example is the original visible light image, and the right is the corresponding segmentation result, which is also the input of the SRSG model. The scene graph generated by the SRSG model is shown in the second and third rows. The second and third rows are different representations of the same scene graphs. The dotted arrows in the figure indicate that the relationship between the two instances has not been correctly predicted. The two-way arrows suggest that the two instances still have an actual semantic relationship after exchanging their positions.

TABLE IV

COMPARISON RESULT OF THE SRSG AND PREVIOUS WORKS IN THE TASK OF GENERATING REMOTE SENSING IMAGE SCENE GRAPHS

Model	mR@50	mR@20
MOTIFS	0.0844	0.0487
IMP	0.0743	0.0555
VCTree	0.1169	0.0764
SRSG-box	0.2216	0.1356
SRSG-seg	**0.2561**	**0.1602**

level segmentation, but the existing methods mostly use the entire image and the bounding box as an input. Therefore, before conducting comparative experiments, we need to generate bounding boxes for objects in the S2SG dataset.

Suppose the coordinates set of the corresponding region of an object in the segmentation result is $\{\boldsymbol{X}, \boldsymbol{Y}\}$. In that case, the coordinate of the upper left corner of the bounding box for the object is $(X_{\min}, Y_{\min})$, and the coordinate of the lower right corner of the bounding box is $(X_{\max}, Y_{\max})$. The bounding boxes are generated for the objects in the S2SG dataset. The predicates between the labeled objects are the same as their corresponding segmentation instances. In this way, the dataset S2SG matches the existing methods. After obtaining the dataset, remote sensing image scene graph generation experiments are conducted on the SRSG model, Neural Motifs [15], Iterative Message Passing (IMP) [14], and VCTree [28]. Because SRSG model directly predicts the predicate between objects on the segmentation results, only the results of predicate classification are compared in the comparative experiment. In the comparative experiment, mR@K is used as the evaluation metric. The experimental results of different models are shown in Table IV. The "SRSG-box" and "SRSG-seg" indicate that the bounding box and pixel-level segmentation are used as the input of the SRSG model, respectively. Comparing the experimental result, it can be seen that the performance of the SRSG model on the dataset S2SG is far better than other methods, and mR@50 is nearly double. This shows that the SRSG model is more suitable for remote sensing image scene graph generation. The SRSG-seg model obtains better performance than the SRSG-box. Using bounding boxes as the input to SRSG-box can only preserve the relative positional information of subjects and objects while omitting their shape features. The decrease in the performance of the SRSG-box indicates that the morphology of the instances is crucial for predicting the semantic relationships between objects in remote sensing images. Therefore, we innovatively detect the relationship between instances in remote sensing images from the pixel-level segmentation results.

The typical distinction of scene graph generation between natural and remote sensing images is that, in the natural scene image, after exchanging the subject and object in the object pairs, the representation obtained is confusing and meaningless, such as ⟨dog-ride-surfboard⟩ → ⟨surfboard-?-dog⟩. However, after exchanging the subject and object in remote sensing images, different predicates can be used to describe the relationship between the interchanged objects: ⟨grass-surround-water⟩ → ⟨water-inlaid-grass⟩. The fundamental reason for this difference is that the distribution patterns of the objects in the two scenes are different, but the SRSG model can capture this characteristic well. For example, the object pairs in Fig. 6(a): ⟨pavement 2-blend-building 2⟩ and

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Fig. 6(b) ⟨vegetation-surround-building 2⟩, after exchanging the position of the two objects, their relationship can be expressed as: ⟨building 2-blend-pavement 2⟩ and ⟨building 2-inlaid-vegetation⟩. Even if the predicate used to describe the relationship changes, the SRSG model can accurately predict the relationship of the objects, indicating that the SRSG model can abstract remote sensing image contents and perform semantic understanding fully.

The remote sensing image scene graphs generated by the SRSG model are shown in Fig. 6. Here, we not only illustrate the scene graphs by nodes and edges (in the third row) but also show them by the pixel-level segmentation and edges (in the second row). The innovative way of illustration can more clearly illustrate the relationship between examples. The scene graph results in Fig. 6 show that even with complex scene relationships, such as "stretch" and "inlaid," the SRSG model has made correct predictions. Even though the scales of different instances in remote sensing images are pretty diverse, it does not affect the accurate prediction of the model, such as ⟨building 1-inlaid-vegetation⟩ in Fig. 6(b). It can be attributed to the input form of the model and the encoding method of the input data. Segmentation is the classification of each pixel in the entire image; that is, the precise labeling of each instance in the image is realized. In this way, the shape of each object in the whole image can be expressed clearly by pixel-level segmentation result. In the SRSG model, different instance shapes are unified by the appropriate coding method. Therefore, the SRSG model with segmentation results as inputs can generate a more complete and accurate remote sensing image scene graph. The scene graphs in Fig. 6 confirm that the relationship between objects in remote sensing images is more dependent on their shapes.

V. Conclusion

In this work, we have analyzed the characteristics of remote sensing images in detail. Because the objects in remote sensing images are highly mixed, we use the pixel-level segmentation results as the inputs of the SRSG model instead of bounding boxes, preventing the labeled objects from containing unnecessary noise, which is a pioneering work. Due to the lack of a dataset for remote sensing image scene graph generation, gmentation results, S2SG. In the dataset S2SG, we ignore the number of the labeled images but focus on the number of each relationship type. With this method, the model can uniformly abstract the characteristics of various relationships. Then, we propose the SRSG model, which takes the coordinates of the objects in the segmentation results as inputs, extracts their morphological features, and finally predicts the predicates between them. In the SRSG model, the positive and negative examples' ratio can be adjusted to make the model discriminate them more clearly. In this way, the object pairs are mapped to their semantic relations. Furthermore, even after changing the position of the subject and object, the SRSG model can still derive the predicate with actual semantic representation. The experimental results demonstrate that the relationships between objects in remote sensing images depend more on their morphological features. In summary, the dataset S2SG and the SRSG model provide novel ideas for generating scene graphs of remote sensing images. The subsequent research will expand the S2SG dataset and combine external knowledge to understand remote sensing images from a higher level.

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